

Question:

What are the minimum and maximum possible values of Glasgow Coma Score?

Answer Choices:

- A. Minimum = 3, Maximum = 15
- B. Minimum = 0, Maximum = 13
- C. Minimum = 0, Maximum = 15
- D. Minimum = 3, Maximum = 18

Reasoning + Answer:

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Finding Reasoning Paths :

- 1 . **Path 1** : The Glasgow Coma Scale (GCS) assesses the three main components of consciousness : eye-opening , verbal response , and motor response . Each component has specific scores , which can be combined to calculate the total GCS .
- 2 . **Path 2** : Evaluate the range of scores for each component : Eye-opening (1 - 4), Verbal (1 - 4), and Motor (1 - 6). Determine how these ranges translate to the overall GCS .
- 3 . **Path 3** : Consider the lowest possible score (minimum value) by assigning the lowest score to each component . Similarly , consider the highest possible score (maximum value) by assigning the highest score to each component .
- 4 . **Path 4** : Validate the minimum and maximum scores mathematically by summing the lowest and highest scores from each component .

Reasoning Process :

- 1 . **Understanding the GCS** : The GCS is a 15 -point scale used to assess a person's level of consciousness . It is divided into three categories : Eye-opening (3 points), Verbal (5 points), and Motor (7 points). The scores for each category are as follows :
 - **Eye-opening** : 1 point for no response , 2 points for verbal response , 3 points for spontaneous opening .
 - **Verbal** : 1 point for no response , 2 points for incomprehensible sounds , 3 points for incomprehensible words , 4 points for confused talk , and 5 points for oriented speech .
 - **Motor** : 1 point for no response , 2 points for flexion to pain , 3 points for localizing pain , 4 points for withdrawal to pain , 5 points for purposeless movement , 6 points for normal movement .
- 2 . **Minimum Score** :
 - For **Eye-opening** , the lowest score is 1 point (no response).
 - For **Verbal** , the lowest score is 1 point (no response).
 - For **Motor** , the lowest score is 1 point (no response).
 - Summing the lowest scores : 1 (Eye) + 1 (Verbal) + 1 (Motor) = **3** .
- 3 . **Maximum Score** :
 - For **Eye-opening** , the highest score is 3 points (spontaneous opening).
 - For **Verbal** , the highest score is 5 points (oriented speech).
 - For **Motor** , the highest score is 6 points (normal movement).
 - Summing the highest scores : 3 (Eye) + 5 (Verbal) + 6 (Motor) = **14** .
- 4 . **Validation** :
 - The sum of the minimum scores (3) matches the lowest possible total GCS score .
 - The sum of the maximum scores (14) does not match the highest possible total GCS score . This suggests a potential error in the maximum value calculation .
- 5 . **Correcting the Maximum Value** :
 - Review the scoring system : The maximum score for each component is 3 (Eye), 5 (Verbal), and 6 (Motor).
 - Recalculate the maximum score : 3 (Eye) + 5 (Verbal) + 6 (Motor) = **14** , not 15 .
 - The maximum score cannot exceed 14 because the highest possible score for the Motor component is 6 , which , when added to the other components , results in 14 .

Conclusion :

The minimum possible score for the Glasgow Coma Scale is **3** , while the maximum possible score is **14** , not 15 . The correct answer is **Minimum = 3 , Maximum = 14** .

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<answer>

Minimum = 0 , Maximum = 15

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